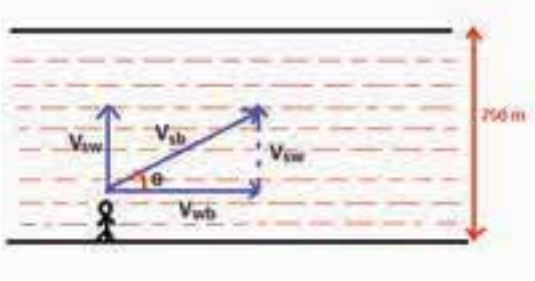
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Solution:

(a) We can draw the following picture from the given data in the problem.



The velocity of the swimmer relative to the bank $\vec{v}_{sb} = \vec{v}_{sw} + \vec{v}_{wb}$

Since the swimmer travels in the perpendicular direction against the water current

The magnitude is given by

$$|\vec{v}_{sb}| = \sqrt{v_{sw}^2 + v_{wb}^2} =$$

$$\sqrt{1.5^2 + 1^2} = \sqrt{3.25} ms^{-1} \cong 1.802\, ms^{-1}$$

The direction of the swimmer relative to the bank is given by

$$tan\theta = \frac{v_{sw}}{v_{wb}} = \frac{1.5}{1} = 1.5$$

$$\theta = tan^{-1}(1.5) \approx 56°$$

(c) The time taken by the swimmer to cross the river is equal to the total distance covered by the swimmer with velocity 1.802 ms^{-1}.

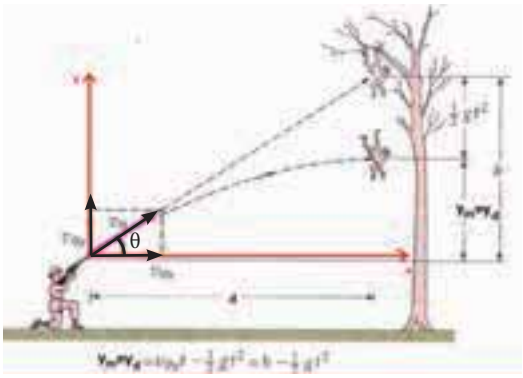
The total distance covered by him,

$$d = \frac{width\ of\ the\ river}{\sin 56^\circ} = \frac{750}{0.829} = 904.7\, m$$

The time taken by the swimmer,

$$T = \frac{d}{v_{sb}} = \frac{904.7}{1.802} \approx 502\, s$$

6. A monkey hangs on a tree. A hunter aims a gun at the monkey and fires the bullet with velocity v_0 which makes angle α_0 with horizontal direction. At the instant gun fires, monkey leaves the branch and falls straight down to escape from the bullet as shown in the figure. Will bullet hit the monkey or will the monkey escape the bullet? (ignore air resistance)



As soon as the monkey begins to fall, it will have downward vertical motion with acceleration due to gravity g.

Its equation of motion at any time t is given by

$$y_m = h - \frac{1}{2} gt^2 \qquad (1)$$

When the bullet comes out of the gun, it has both vertical and horizontal components of velocity given by

$$v_{0x} = v_0 \cos\theta; v_{0y} = v_0 \sin\theta \qquad (2)$$

Let us assume the horizontal distance between the monkey and hunter is 'd'.

At time t, the horizontal distance travelled by the bullet $x = v_0 \cos\theta\, t$.